

KRUNAL SOLANKI

Senior .NET Architect | Azure Solutions Architect | Microservices & Cloud-Native Specialist

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PROFESSIONAL SUMMARY

Senior .NET Architect with 18+ years delivering enterprise-grade microservices, cloud-native APIs, and distributed systems on Azure. Deep expertise in ASP.NET Core, .NET 6/8/9, Clean Architecture, SOLID principles, CQRS, DDD, and design patterns (Builder, Factory, Visitor, SAGA). Extensive hands-on experience integrating Azure APIM, Service Bus, Cosmos DB, Key Vault, Redis, and Event Hub. Proven track record implementing OAuth2/JWT security, Polly resilience, gRPC/REST/GraphQL APIs, Entity Framework, and CI/CD pipelines—leading distributed teams and delivering platforms serving 49,000+ monthly users.

CORE TECHNICAL SKILLS

.NET Stack	.NET 6/8/9/10, ASP.NET Core, C#, Entity Framework Core, LINQ, gRPC, REST, GraphQL, Web API
Design Patterns	SOLID, Clean Architecture, CQRS, DDD, Builder, Factory, Visitor, SAGA, Repository, Mediator, Polly Resilience
Azure Services	APIM, App Gateway, App Configuration, Key Vault, Service Bus, Event Hub, Cosmos DB, Azure SQL, Functions, AKS, Blob Storage, Azure Monitor, App Insights
Security & Auth	OAuth2, JWT, SAML, ASP.NET Policy Framework, Azure Entra ID (B2B/B2C), Managed Identities, SSL/TLS
Microservices	Microservices Architecture, DAPR Runtime, Docker, Kubernetes (AKS), Event-Driven Architecture, Distributed Caching (Redis)
Async & Performance	Async/Await, Task Parallel Library, Buffering/Streaming, Serialization Frameworks, Distributed Tracing
Testing & Quality	NUnit, MSTest, xUnit, Swagger, SonarQube, Veracode, Checkmarx, TDD, 90%+ Code Coverage
CI/CD & DevOps	Azure DevOps, GitHub Actions, GitLab CI, Docker, Kubernetes, Terraform, ARM Templates, Blue-Green Deployments
Databases	Azure SQL, Cosmos DB, SQL Server, Redis Cache, MongoDB, Entity Framework Core, ADO.NET, Stored Procedures
Messaging	Azure Service Bus, Event Hub, RabbitMQ, Kafka, Publish-Subscribe Patterns

AZURE CERTIFICATIONS

Microsoft Certified: Azure Solutions Architect Expert [AZ-304]

Azure AI Engineer Associate [AI-102]

Azure Administrator [AZ-104]

AWS Certified Architect

Azure Data Fundamentals

Azure Fundamentals

DOMAIN EXPERTISE

Financial Services (10 years) • Title Insurance & Mortgage • Technology / SaaS • Aviation (Emirates, Qatar Airways) • Supply Chain (Unilever) • Payments (Visa/PCI-DSS) • Pharmaceutical (Pfizer) • Education Technology • E-Commerce

PROFESSIONAL EXPERIENCE

DELL TECHNOLOGIES | Senior Software Architect & AI Innovation Lead *April 2021 – Present*

ISG Marketing Platform Modernization — .NET 9 Web API, ASP.NET Core, CosmosDB, Redis, Azure Service Bus, APIM, AKS

- Architected cloud-native microservices platform on ASP.NET Core using Clean Architecture, CQRS, and SOLID principles serving 49,000+ monthly users with 99.9% availability.
- Designed event-driven architecture using Azure Service Bus (publish-subscribe) to maintain data consistency handling 10,000+ transactions/minute across distributed microservices.
- Integrated Azure APIM as API gateway with rate limiting, JWT/OAuth2 authentication policies, and API versioning for secure, governed API access.
- Implemented distributed caching with Redis, reducing database load by 75% and enabling sub-second response times for high-traffic catalogue services.
- Built Polly-based resilience policies (retry, circuit breaker, timeout) for all inter-service HTTP and Azure Service Bus communications.
- Implemented Entity Framework Core with Repository and Unit-of-Work patterns; optimized complex LINQ queries and stored procedures against Azure SQL and CosmosDB.
- Established CI/CD pipelines with Azure DevOps and GitLab CI including SonarQube/Veracode quality gates, reducing deployment time by 75% and improving code coverage to 90%.
- Secured all API endpoints using ASP.NET Core Policy Framework, OAuth2/JWT, and Azure Key Vault for secret management and certificate rotation.
- Documented all APIs with OpenAPI 3 / Swagger, enabling rapid consumer onboarding and automated contract testing.

Dell Masthead Search — AI Agent — .NET 8 Web API, Azure OpenAI, Redis, CosmosDB, Application Insights

- Designed and led .NET 8 microservice exposing gRPC and REST endpoints for AI-powered search, implementing async/await and IHostedService background workers for high throughput.
- Implemented distributed caching (Redis) and token-optimization middleware achieving 40% cost reduction and sub-second latency for 10,000+ daily queries.
- Built IoC container configuration using .NET DI framework with scoped, transient, and singleton lifetimes following ASP.NET Core best practices.
- Integrated Azure Application Insights for distributed tracing, custom telemetry, and performance monitoring dashboards across microservice boundaries.

FIRST AMERICAN | Senior .NET Technical Lead

January 2011 – April 2021

Technical lead for enterprise title insurance and real estate applications—10 years of domain expertise aligned with insurance industry requirements.

AttorneyFirst — Title Workflow Platform — ASP.NET Core, .NET, Azure Cloud, Azure SQL, Azure Storage, OAuth2

- Architected secure B2B workflow platform using ASP.NET Core with OAuth2/SAML authentication, Azure Entra ID integration, and role-based authorization policies.
- Designed document management system using Azure Blob Storage with lifecycle policies and tiering, achieving 25% storage cost reduction for GB-scale monthly volumes.
- Built flexible search and template engine using Entity Framework Core with LINQ; implemented async stored procedures improving document retrieval efficiency by 40%.
- Implemented exception handling middleware and structured logging aligned with ASP.NET Core best practices, enabling rapid root-cause analysis.

FirstAM EDU — Multi-Tenant Educational Platform — Angular, .NET Core, Azure Functions, Azure SQL, Redis, Azure Blob

- Designed multi-tenant architecture with comprehensive RBAC using ASP.NET Core Policy Framework, Azure SQL row-level security, and encrypted Azure Key Vault secrets.
- Implemented Redis cache-aside pattern and SQL query optimisation, achieving 45% faster page loads and 60% quicker API response times under 15,000 concurrent users.
- Developed Azure Functions (serverless microservices) for automated certificate generation and background processing with Polly retry policies for transient fault handling.

- ❑ Wrote NUnit/MSTest unit and integration test suites with Moq achieving 90%+ code coverage; documented APIs with Swagger..

FAMOS — Mortgage Operations System — C#, .NET Framework, ASP.NET, ADO.NET, SQL Server, NUnit

- ❑ Engineered rule engine with dynamic routing logic using C# design patterns (Strategy, Factory) reducing file assignment time by 65%.
- ❑ Developed OCR-integrated document processing pipeline with automated classification reducing errors by 75% and improving accuracy to 98%.
- ❑ Implemented stored procedures and ADO.NET data access layer with optimised bulk operations for high-volume mortgage transaction processing.

DXC TECHNOLOGIES (formerly CSC India) | Senior Software Engineer*August 2005 – December 2010*

- ❑ Built enterprise ASP.NET / WCF applications for Fortune 500 clients across aviation (Emirates, Qatar Airways), supply chain (Unilever), **payments (Visa)**, and pharma (Pfizer).
- ❑ Developed RESTful and SOAP service integrations with secure authentication (OAuth, SSL/TLS) and PCI DSS-compliant payment gateway modules.
- ❑ Implemented layered architecture and OOP design patterns in .NET Framework, establishing coding standards adopted across multiple client engagements.

EDUCATION

Bachelor of Engineering in Information Technology | C U Shah College of Engineering & Technology

KEY ACHIEVEMENTS

- ❑ 3× Dell Inspire Award recipient for technical excellence and platform innovation.
- ❑ Reduced manual testing effort by 90% via multi-agent AI test orchestration system built on .NET Core and Azure Functions.
- ❑ Achieved 99.9% platform availability and near-zero downtime through Blue-Green deployment pipelines with Azure DevOps.
- ❑ Managed 15+ engineers across 4 continents, increasing sprint velocity by 25% through standardised Agile and Clean Architecture practices.
- ❑ Delivered 70% faster microservice onboarding across 10+ internal teams via a centralised CI/CD framework with consistent quality gates.